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Enhancing Teacher Support in Learning Technologies: A Human-Centered Approach with WearableLearning

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Abstract. Learning technologies are widely used in K-12 classrooms but often fall short in meeting teachers' needs. This study employs a multi-method, human-centered approach to design teacher support for WearableLearning (WL), a learning technology for math and computational thinking through embodied learning. We began with a survey of 52 K-12 teachers to assess how the learning technologies they use in their classrooms support or fail to support them. We identified eight areas where learning technologies offer support and five areas where they lack support, and used these insights to inform the design of a preliminary teacher dashboard for WL. We conducted two rounds of interviews with three teachers to refine the design through iterative feedback. Finally, we administered a questionnaire to eight pre-service teachers to further explore teacher needs. This process resulted in a low-fidelity teacher dashboard prototype, which will undergo cognitive walkthroughs in future work. The findings from our study can be used to better design teacher support in learning technologies.

Keywords: Teacher dashboards, learning technologies, human-centered design.

1 Introduction

Learning technologies are playing an increasingly central role in K-12 education, promising to enhance both student learning and teacher instruction [1]. However, while these technologies aim to improve education, teachers often struggle to use them due to insufficient training and understanding of how to apply them effectively [2]. Furthermore, many technologies fail to adequately support teachers in monitoring students and understanding students' processes, misconceptions, and knowledge gaps [3].

In recent years, there has been growing interest in enhancing teacher support through dashboards [3, 4]. Teacher dashboards empower teachers with real-time student data, enabling deeper insights, reflection, and data-driven decision-making [3, 5]. While prior work has focused on teacher dashboards for traditional learning technologies (*i.e.*, systems where students remain seated at desks) [3, 4, 5, 6, 7], teacher dashboards for less traditional technologies that encourage physical movement remain underexplored.

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One such system is WearableLearning (WL) [8, 9], which promotes learning through embodied gameplay, blending the physical classroom with the digital space.

While prior work has focused on professional development (PD) to support teachers adopting WL [10], no work has explored supporting teachers in real-time while students play WL games. Questions remain regarding how teachers can monitor students and view their progress and activity. To address this gap, we used a multi-method, human-centered approach to build teacher support for WL. In doing so, we address the following research questions: *How do learning technologies support or fail to support teachers?* and *How can teacher support be integrated into WL to best meet teachers' needs?*

2 Background

2.1 Challenges of Learning Technologies in the Classroom

Learning technology is a broad term that encompasses any instructional tools designed to aid student learning [2, 11]. The pace at which these technologies are developed often outpaces the ability of teachers to learn how to use them effectively [2]. This is partly due to the Gartner hype cycle, which describes how new technologies surge in adoption due to excitement but decline when expectations aren't met [2, 12, 13]. When teachers try to incorporate new technologies but encounter challenges early on, they may abandon the technology and become apprehensive about trying new ones [2].

One way to address these challenges is through PD [14, 15]. Ertmer and Ottenbreit-Leftwich (2010) argue to support "21st-century teaching and learning, we need to help teachers understand how to use technology to facilitate meaningful learning." While PD can help teachers, for it to be effective, it must demonstrate the technology is easy to use and beneficial for instruction, which is often not achieved in PD models [15].

PD may help teachers adopt a technology, but supporting teachers *while* students use it is another challenge. Teachers need real-time, actionable insights to support their awareness, reflection, and decision-making [3, 16]. More specifically, in a study on real-time learning analytics, Holstein *et al.* (2017) identified seven needs of teachers from learning technologies: (1) seeing students' thought processes, (2) identifying who is stuck or (3) "almost there," (4) supporting multiple students simultaneously, (5) monitoring misbehavior, (6) detecting misconceptions, and (7) recognizing patterns of careless errors [3]. These needs are often met through real-time teacher dashboards [4].

2.2 Dashboards for Supporting Teachers

Dashboards empower teachers with real-time student data, enabling deeper insights, reflection, and data-driven decision-making [3, 5]. Specifically, Poh *et al.* (2023) identified seven key tasks supported by teacher dashboards: (1) viewing data at student, group, and class levels, (2) viewing students' task performance, (3) providing feedback and support to students, (4) personalizing learning, (5) monitoring and supporting collaboration, (6) orchestrating activities, and (7) reflecting after activities.

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Teacher dashboards support these tasks through simple visualizations, symbols, colors, and the ability to adjust data granularity [4, 5, 6, 7]. As Molenaar and Knoop-van Campen (2017) noted in regard to the *distributed cognition theory*, "instruments can indeed support professionals, when these instruments fit seamlessly into the activities of a professional" [5, 17]. That is, effective teacher dashboards must align with teachers' tasks to provide meaningful support, and features like simple visualizations and colors best align with the primary task of monitoring and viewing student data [4, 5].

AI integration in educational technologies has grown in recent years [18, 19, 20, 21, 22]. As teacher dashboards become increasingly AI-powered, questions remain about teachers' willingness to adopt and trust AI. Through discussions with teachers, we explore their perspectives on AI support, discussed in *Section 4: Results and Discussion*.

2.3 WearableLearning

WL is a learning technology that promotes math and computational thinking through *game creation* and *gameplay* [8, 9]. Unlike traditional technologies that confine students to desks, WL creates a blended learning environment by integrating physical and digital spaces, supporting active learning through movement [9, 10, 23].

Games in WL are created using a simplified drag-and-drop programming language based on finite state machines (FSM). An FSM is an abstract model that represents a system's behavior (in this case, a game) through two main components: *states* and *transitions*. While prior research has focused on how students create games [9, 23, 24], this paper focuses on game playing, specifically on supporting teachers while students play.

One example of an embodied WL game is *Tangrams Race*, a geometric relay game where students race across the classroom to collect tangram pieces based on clues displayed on their armband-worn devices. Other examples include stepping forward and backward on a large number line, scavenger hunts, and dance-based activities.

Compared to traditional learning technologies, WL presents greater monitoring challenges since students are moving around and the smaller, wearable devices are harder to monitor. A key limitation of the current WL platform is the lack of effective teacher support during gameplay [24]. To address this gap, this study explores ways to better support teachers while students play games by first examining how other technologies provide support and then exploring how to integrate this into WL.

3 Methodology

To understand how to best support teachers within the WL platform, we used a multi-method, human-centered design approach with three phases.

3.1 Survey

We recruited teachers via publicly listed emails on school district websites. 52 teachers (11 primary, 16 middle, 25 high school) from diverse public and private schools in the Northeastern US participated. The survey included five open-ended questions (*e.g.*,

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What learning technologies support you and how?). We analyzed responses with word clouds and affinity diagramming to identify areas of support and lack thereof.

3.2 Interviews

Based on the survey findings, we designed an initial mockup of a teacher dashboard for WL, which guided our semi-structured interviews. We recruited two middle school teachers from a prior WL PD study, neither of whom had participated in the survey. Additionally, we recruited a primary school teacher who was a member of the WL research team at the time but whose research focused on creating WL games rather than platform design or development. They had no apparent stake in a dashboard for WL.

The first interview round explored teachers' experiences with WL and their perspectives on how it supports them. These 45–60-minute Zoom interviews were conducted individually and recorded with consent. While discussions were teacher-driven, we prepared guiding questions such as *In what ways does WL currently support/not support you?* In the second half, we presented our initial teacher dashboard mockup and asked for feedback, including *What data is missing that you wish you could see?* Using grounded theory, we gathered insights from our discussions to revise the mockup.

The second interview round focused on teachers' perspectives on the revised mockup. These 30-minute Zoom interviews were conducted individually and recorded with consent. As in the first round, discussions were teacher-driven, with guiding questions like *What information do you see here?* and *How would you use this information in your classroom?* Grounded theory was used again to inform a low-fidelity prototype.

3.3 Questionnaire

In recognition of our small interview sample, we administered a questionnaire to eight pre-service teachers in a Master of Arts in Teaching program. Before responding, participants played *Tangrams Race*. The questionnaire reflected themes from the interviews, including student monitoring, a common concern among interviewees.

4 Results and Discussion

From the survey, teachers cited 108 learning technologies, with Google technologies (*e.g.*, Chromebooks, Google Classroom) being the most frequently mentioned. Through affinity diagramming, we identified eight areas where learning technologies *offer support* (see Table 1) and five areas they *lack support* (see Table 2) for teachers. While most teachers' needs identified by Holstein *et al.* (2017) are addressed to some capacity by the *offer support* areas (*e.g.*, through assessment reports), an alarming issue is the lack of support for the need of student monitoring. Based on our findings, we identified two priority areas for our mockup: *assessment and feedback* and *student monitoring*.

The top of Fig. 1 shows two key views from our initial dashboard mockup: *Class* and *Individual*. The *Class* view displays overall student progress, including each student's placement and performance on a simplified FSM of the game (*e.g.*, red circle

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represents a struggling student). The *Individual* view shows more detailed performance data for a single student, along with AI-generated suggestions for assisting them. Together, these views aim to support teachers in viewing both class and student data (*e.g.,* task performance) and real-time support and monitoring, addressing three of the seven key teacher tasks identified by Poh *et al.* (2023). (Although not shown in Fig. 1, our mockup also includes *Orchestrate, Task,* and *Report* views for the remaining tasks.)

Table 1. Areas where learning technologies *offer support* to teachers.

Area	Sample Response
Instruction and Content Delivery	"I use [it] to get quick reports that I can use to guide instruction."
Personalized Learning	"[It] allows me to assign and differentiate work for students."
Engagement and Interactive Learning	"[They] allow me to be able to post interactive websites or work online."
Assessment and Feedback	"They allow me to see student growth, performance data, and student choice instantaneously."
Classroom Organization	"[It] definitely supports me by allowing me to keep my classroom and gradebook much more organized."
Communication and Collaboration	"[It] allows me to tutor students if they can't be in school, talk to parents who can't come in for meetings, or teach virtually."
Efficiency and Time Management	"Anything that can simplify the work that I have to do behind the scenes, improves the work that my students see."
Technology Integration	"[They] support me as a teacher because I have the ability to access various platforms with my students."

Table 2. Areas where learning technologies *lack support* for teachers.

Area	Sample Response
Availability	"We need more [learning technologies] that are accessible and free and teachers do not have to constantly search for them."
Usability	"Some are so busy that it is hard to quickly do what you want to."
Customizability	"[M]ore customizable interfaces so that I can construct my own unique interactives and activities."
Student Monitoring	"I wish we could monitor more easily. We use [monitoring software] but its multiple steps and takes up the whole screen and then becomes another "tab" to monitor."
Professional Development	"I wish that I received more [PD] on using more platforms. I feel like there [are] more websites and platforms to use but finding the time to implement them with lack of training can be a challenge."

Our first round of interviews revealed that our dashboard mockup needed refinement to better support teachers. Teachers emphasized the importance of seeing the *exact* screens of students. While they could view progress in the initial mockup, they couldn't always understand the exact challenges students faced (*e.g., Why hasn't a student progressed?*). Teachers also preferred the *exact* FSM over the simplified version. Initially,

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we simplified the FSM based on research suggesting that teachers prefer simple visualizations [4]. However, our findings highlighted an important counterpart to simplicity: *familiarity*. Teachers expressed that the exact FSM was simpler because they were familiar with it from creating games. This aligns with the *familiarity heuristic* in which people subjectively prefer what they recognize [25]. In response, we made two key changes to the mockup (bottom of Fig. 1): a live view of students' screens and an exact FSM. In the second round of interviews, teachers strongly expressed the benefits of these changes for monitoring. Pre-service teachers also echoed this, with one also suggesting a student device pause feature that we added to the low-fidelity prototype.

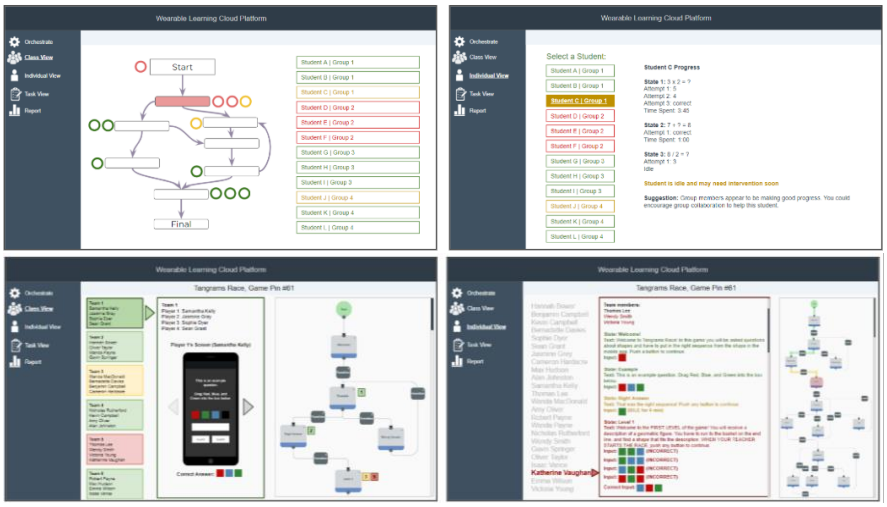


Fig. 1. Initial mockup *Class* (top-left) and *Individual* (top-right) views along with the revised *Class* (bottom-left) and *Individual* (bottom-right) views.

We also discussed AI-generated suggestions with interviewed teachers. In the first round, one teacher was apprehensive to use them over their own inferences, while the other found them too verbose. In response, we removed the AI suggestions, and in the second round, all teachers preferred seeing more student data over AI suggestions. While our study suggests preliminary results towards teachers not having interest in AI-generated support at the moment, we wish to explore this more in the future.

5 Conclusion

In this study, we used a multi-method, human-centered approach to design a teacher dashboard. Our analysis identified eight areas where learning technologies offer support and five where they lack support for teachers. Additionally, our results suggest the importance of showing students' exact screens and designing with teacher familiarity in mind. Regarding AI support, our results suggest teachers have limited interest at the moment. Future work will evaluate our prototype through cognitive walkthroughs.

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